2.7 Fluid mechanics	**2.7.1**	**Knowledge, understanding and application of aerodynamics and hydrodynamics to appropriate sports contexts.**
	2.7.2	**Factors affecting fluid friction and air resistance: velocity, drag force, mass, streamlining and surface characteristics of body.**
	2.7.3	**Interaction of lift forces with objects: upward and downward lift forces, angle of attack and the Bernoulli effect.**
	2.7.4	**Types of spin: topspin, backspin, sidespin. Magnus effect and how they impact on flight path and bounce.**
	2.7.5	**Principles of fluid mechanics and how it has influenced technological advancements in technique modification, clothing/suits, equipment/apparatus.**

Assessment information

- First assessment: May/June 2018.
- The assessment is 2 hours and 30 minutes.
- The assessment comprises two sections: Section A – Applied anatomy and physiology and Section B – Exercise physiology and applied movement analysis.
- The assessment is out of 140 marks.
- Students must answer all questions.
- The assessment consists of short-answer, long-answer and extended-answer questions.
- Calculators may be used in the examination. Information on the use of calculators during the examinations for this qualification can be found in *Appendix 8: Calculators*.

Sample assessment materials

A sample paper and mark scheme for this component can be found in the *Pearson Edexcel Level 3 Advanced GCE in Physical Education* Sample Assessment Materials (SAMs) document.